## SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

### Department of Computer Science and Engineering

### ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

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|---|---|---|---|
| **Course Code:** | CSA06 | **Course Title:** | Design and Analysis of Algorithms |
| **CO Assessed:** | CO1 – Precision (BL1, BL2, BL3) | **Assessment:** | Assessment Tool 1 – Application Based Questions |
| **Weightage:** | 40% of CIA | **Total Marks:** | 100 (2 Questions × 50 Marks) |
| **CO1 Statement:** | *Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)* | | |
| **Student Name:** | Haresh Kumar N L | **Reg. No.:** | 192425009 |
| **Section / Batch:** | AI&ML | **Date:** | 16/07/2026 |

**Instructions to Students**

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

| Question Type | Focus Area | Questions | Marks |
|---|---|---|---|
| **Application-Based Questions** | Apply Master Theorem and asymptotic analysis to real-world scenarios | Q1 – Q2 | Q1 –50<br>Q2 –50 |
| | **GRAND TOTAL:** | **100 Marks** | |

### SECTION A – APPLICATION BASED QUESTIONS

**Q1. Image Processing System**

An image processing algorithm divides images recursively: $T(n) = 4T(n/2) + n$. Apply the Master Theorem to find the time complexity. Clearly identify parameters and case selection. Analyze how performance changes with image size.

**Q2**. File Sorting in Distributed Systems

A distributed sorting algorithm follows: $T(n) = 2T(n/2) + n \log n$. Apply the Master Theorem and determine the complexity. Justify the selected case. Discuss efficiency in handling large-scale distributed data.

**Code of Conduct**

*I* **Haresh Kumar N L (192425009)** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ___________________________

**ANSWER:**

**Q1. Given:** $T(n) = 4T(n/2) + n$

**Apply the Master Theorem.**

**Step 1: Identify Parameters**

**The standard form is:** $T(n) = aT(n/b) + f(n)$

**Compare:**

**a = 4 , b = 2 , f(n) = n**

**Step 2: Compute**

$$n^{\log_b a} = n^{\log_2 4} = n^2$$

**Since**

$$f(n) = n = O(n^{2-1}) = O(n^1)$$

**which is polynomially smaller than** $n^2$**,**

$$f(n) = O(n^{\log_b a - \epsilon})$$

**where** $\epsilon = 1$**.**

**Therefore, Master Theorem Case 1 applies.**

**Step 3: Time Complexity**

$$T(n) = \Theta(n^{\log_2 4}) = \Theta(n^2)$$

**Analysis**

- **The recursive calls dominate the running time.**
- **As image size increases, execution time grows proportionally to** $n^2$**.**
- **The algorithm performs well for small images but requires significantly more processing for larger images due to quadratic growth.**

**Final Answer**

- **a = 4**
- **b = 2**
- **f(n) = n**
- **Master Theorem Case: Case 1**
- **Time Complexity:** $\boxed{\Theta(n^2)}$

**Q2. Given:**

$$T(n) = 2T(n/2) + n\log n$$

**Apply the Master Theorem.**

**Step 1: Identify Parameters**

**Compare with**

$$T(n) = aT(n/b) + f(n)$$

- **a = 2**
- **b = 2**
- **f(n)=n\log n**

**Step 2: Compute**

$$n^{\log_b a} = n^{\log_2 2} = n$$

**Now compare:**

$$f(n) = n\log n = \Theta(n\log n)$$

**This matches**

$$\Theta(n^{\log_b a}\log^1 n)$$

**Hence Master Theorem Case 2 applies.**

**Step 3: Time Complexity**

**For Case 2,**

$$T(n) = \Theta(n\log^{k+1} n)$$

**where $k = 1$.**

**Therefore,**

$$T(n) = \Theta(n\log^2 n)$$

**Efficiency in Handling Large-Scale Distributed Data**

- **The algorithm divides the data evenly across multiple systems.**
- **Each node sorts its local data efficiently.**
- **The additional $n\log n$term represents communication and merging overhead.**
- **Overall complexity is $\Theta(n\log^2 n)$, making it efficient and scalable for large distributed datasets, though communication costs increase with data size.**

### RUBRICS FOR CALCULATION

| | | | | | |
|---|---|---|---|---|---|
| **Understanding of Problem Context** | **20%** | **Clearly understands the problem and accurately identifies all requirements** | **Good understanding with minor gaps** | **Basic understanding; some requirements unclear** | **Poor understanding or misinterpretation of the problem** |
| **Application of Concepts** | **30%** | **Accurately applies appropriate concepts to solve complex problems** | **Applies relevant concepts correctly with minor errors** | **Applies basic concepts with limited accuracy** | **Fails to apply appropriate concepts** |
| **Problem-Solving Approach** | **20%** | **Logical, well-structured, and efficient approach** | **Mostly logical approach with minor gaps** | **Basic approach with some inconsistencies** | **Illogical or unclear approach** |
| **Accuracy of Solution** | **20%** | **Completely correct solution with proper justification** | **Mostly correct with minor errors** | **Partially correct solution** | **Incorrect or incomplete solution** |
| **Clarity** | **10%** | **Clear, well-organized, and properly explained solution** | **Understandable with minor clarity issues** | **Limited clarity and explanation** | **Poorly presented and difficult to understand** |

**Marks Evaluation:**

| Question No. | Understanding of Problem Context (20%) | Application of Concepts (30%) | Problem-Solving Approach (20%) | Accuracy of Solution (20%) | Clarity (10%) | Total Marks |
|---|---|---|---|---|---|---|
| Q1<br>(50 Marks) | | | | | | |
| Q2<br>(50 Marks) | | | | | | |
| Overall Total (100) | | | | | | |

| Faculty In-charge | Course Coordinator | Head of Department |
|---|---|---|
| | | |
| Signature: ______ | Signature: ______ | Signature: ______ |
| Date: ____ | Date: ____ | Date: ____ |